Probability and Statistics B - Practical Sheet 4

Getting started:

1. Open **RStudio** (do not use the base R application).
2. From the menu, select **File** → **New File** → **R Script**. This will create a blank script where you can type and run your R commands.
3. Immediately save your script with a meaningful name, e.g., `Practical1.R`, using **File** → **Save As**. Choose a location you can easily find later.
4. Type your commands in the script and run them using:
   - **Ctrl + Enter** (Windows) or
   - **Cmd + Enter** (Mac)
5. When you have your answers, copy the relevant output (not the entire script) into a Word document.

Important: Always keep your R script saved. This is good practice and ensures you have a record of your work.

1. The data below refers to violent crime rates for a random sample of 20 American areas in 2013 and 2015. The same areas are used both years so the samples are paired. The values show the number of reported crimes per 100,000 population.

   `y2013:`

   ```
   613.8 453.1 826.7 322.2 640.9 807.1 716.7 190.9 217.7 125.1
   553.1 914.1 409.6 53.7 487.8 342.8 120 171.8 190.9 237.2
   ```

   `y2015:`

   ```
   610.5 571.2 1157.2 485.5 934.6 950.1 720.1 288 298.6 75.7
   575.8 1072.1 631 33.3 460.2 377 144.5 161.5 225.7 269.5
   ```

   Obtain a 99% confidence interval for the difference in the population means between the two years. Discuss your result; including any assumptions you need for the test to be valid. (Use `t.test` with the subcommands `conf.level = 0.99, paired=T`). How does this compare with the result if the samples had not been paired?

2. A set of data consists of the following 15 values. Suppose that the data correspond to a random sample from a population.

   ```
   12.4 16.7 20.5 17.0 10.6 14.9 16.8 21.6 17.7 10.2 15.1 18.2 19.3 21.0 15.8
   ```

   (a) Obtain a 95% confidence interval for the underlying population mean.

   (b) Test $H_0 : \mu = 18.5$ vs $H_1 : \mu \neq 18.5$:

   ```
   t.test(x, alternative = "two.sided", mu = 18.5)
   ```

   Mention the t statistic, p-value, and interpret them.

3. Repeat Question 2(b) for the following hypotheses:

   (a) $H_0 : \mu = 15$ vs $H_1 : \mu > 15$

   (b) $H_0 : \mu = 14$ vs $H_1 : \mu > 14$

(c) $H_0 : \mu = 13$ vs $H_1 : \mu > 13$

Use:

```
t.test(x, alternative = "greater", mu = ...)
```

4. Perform a two-sided hypothesis test to determine if these data come from a population with mean 260 using $\alpha = 0.05$:

```
241 251 258 246 202 246 246 301 260 228 275 266 264 250 258 265 240 227 271 306
269 252 242 274 222 244 218 228 247 202 287 279 242 262 268 226 303 214 242 233
```

5. The identity of witnesses in sensitive court cases is often protected using various camera techniques, but it is hoped that this will not influence the outcome of a trial. One useful variable to be measured is the number of facts correctly remembered by a random sample of participants who watched a filmed interview featuring the witness under two different masking conditions "pixilation" or "negative image". The results are as follows:

```
Pixilation:

68 75 65 69 70 72 65 66 58 62

Negative Image:

59 61 62 57 64 57 60 61 51 55
```

Conduct a hypothesis test to investigate whether more facts are remembered by populations who view the witness by "pixilation" than by those who view the witness by "negative image". A useful R command is:

```
t.test(x, y, var.equal = TRUE, alternative = "greater")
```

(or `alternative = "less "` depending on how you define the differences).

6. In a study of arachnophobia (fear of spiders) the anxiety levels of twelve subjects are measured when they are asked to handle a jar containing a real spider and again (on a different occasion) when they are asked to handle a card with a picture of a real spider on it. The resulting data are:

```
Subject:  1  2  3  4  5  6  7  8  9  10 11 12
Real:    40 35 50 55 65 55 50 35 30 50 60 40
Picture: 30 35 45 40 50 35 55 25 30 45 40 50
```

Perform a suitable t-test using R to investigate whether real spiders evoke greater levels of anxiety as compared to pictures of real spiders.